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Pulse voltage operation of two-to-four-phase voltage-induction-type electrostatic motor

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Abstract. This paper analyzes pulse voltage operations of a two-to-four phase voltage-induction-type electrostatic motor (VITEM) to clarify the appropriate driving sequences. The paper also evaluates transient charge effects that can possibly enhance the thrust force. VITEM has been studied as a novel centimeter-sized electrostatic film motor whose power to the slider is supplied indirectly through its induction electrodes. Since a smooth synchronous driving was aimed in the past studies, high-frequency AC voltages have been used for the driving. This paper demonstrates that straightforward usage of simple pulse sequences cannot realize continuous stepping driving, and proposes several different sets of modified pulse sequences to realize continuous stepping driving using low-switching-rate pulse voltages. In the experiments, the motor showed continuous stepwise driving with a step width of about half of the electrode pitch. In addition, transient characteristics of charges on films were evaluated experimentally under the pulse driving condition, and contribution for thrust force enhancement was clarified.

Keywords: Pulse voltage, electrostatic motor, two-to-four phase, voltage induction

1. Introduction

Electrostatic motors have succeeded first in the MEMS field in 1980s [1,2], and since then, various types of motors have been studied [3–13]. The success of MEMS electrostatic motors sparked development of macroscale electrostatic actuators that can generate sufficient force/torque for macroscale applications [14,15]. Such actuators can be realized by integrating a large amount of small electrodes onto large-size thin substrates, such as plastic films [14,15].

One major classification of those macroscale motors is based on the existence of the electrodes on the slider. Whereas having energized electrodes in a slider contributes to the superior output performances [15] and fine controllability as a synchronous motor, electrode-free configurations for the slider can realize non-tethered operations that can eliminate mechanical disturbances caused by voltage feeding cables [14].

Recently, electrostatic motors of an intermediate type, which have energized electrodes in a slider but free from cabling, have been proposed [16,17]. These motors realize high voltages on slider electrodes by electrostatic induction and they differ in their methods for high voltage induction. A voltage-induction-type electrostatic motor (VITEM) [16,18,19] energizes slider's electrodes by electrostatic induction through dedicated built-in capacitors, and operates as a synchronous motor, whose speed is proportional to the difference of frequencies of applied voltages. A resonance-type induction motor [17]

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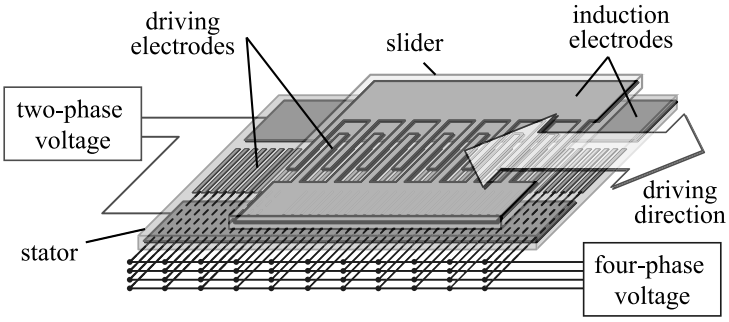


Fig. 1. Two-to-four phase voltage-induction-type electrostatic motor. A slider and a stator have two-phase and four-phase parallel driving electrodes, respectively. Induction electrodes allow voltage applications to the slider through capacitive connections.

utilizes LC resonance with external inductors to enhance the induced voltages, and its resultant motion becomes asynchronous like electromagnetic induction motors [20–22]. These motors are free from disturbances caused by cabling, while maintaining relatively large output performances. This paper focuses on the former type, VITEM, since it is attractive due to no need for external elements for high voltage induction, which leads to a simpler motor structure.

Original VITEM [16] was proposed to have two-phase slider electrodes, for space efficiency and ease of fabrication, and four-phase stator electrodes. This configuration was referred to as "two-to-four phase" configuration [19]. In the previous studies, the two-to-four phase VITEM has been operated using high-frequency ac voltages to eliminate undesired vibrations that originate in the nature of two-phase electrodes [16,19]. This paper analyzes motor operation under low-switching-rate pulse voltages, since use of such pulse voltages can ease the motor operations and widen the application areas of the VITEM. The use of low-switching-rate pulse voltages, however, cannot be realized in a straightforward way; a simple pulse sequences can result in halt of motor operations, as will be demonstrated in this paper. This paper discusses the cause of the motion failure under simple pulse sequences and proposes several sets of modified pulse voltage sequences that can realize continuous stepping operations.

In the next section, the paper theoretically analyzes thrust force characteristics under a simple pulse voltage sequence to provide theoretical basis for the discussions. The section also shows that a simple pulse sequence cannot realize continuous stepping operations. Then, section III proposes modified sequences to realize continuous motor operations, which are experimentally verified in section IV. Section IV also reports on increases of induced voltages after constant voltage applications and discusses the possibility of thrust force enhancement by this effect.

2. Basic analysis for pulse voltage operation

2.1. Basic structure of the motor

Figure 1 shows a schematic diagram of a voltage-induction-type motor of the two-to-four phase configuration. The motor consists of two films that work as a stator and a slider respectively. Both films are equipped with parallel multi-phase electrodes for driving and planar electrodes for voltage induction, both of which are covered with an insulating layer. The stator's driving electrodes have a four-phase structure and are aligned with a regular pitch of p. The slider's electrodes have a two-phase structure and are aligned with double pitch of the stator electrodes, $2p$. As a result, both of the stator and slider

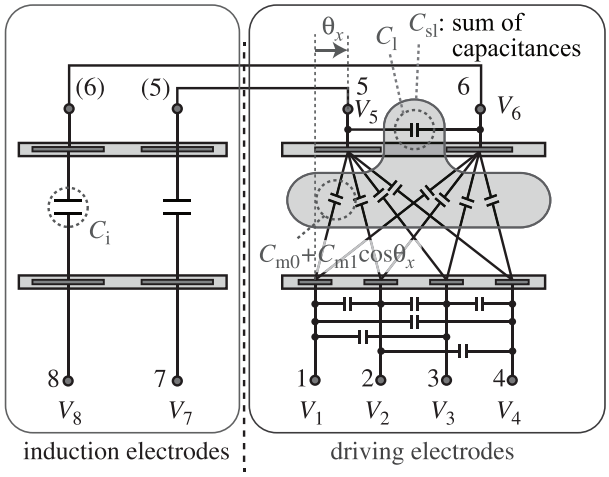


Fig. 2. Eight-terminal capacitance network model for two-to-four phase VITEM.

electrodes repeat with the same cycle, hereinafter called structural period, which is $4p$. Besides the driving electrodes, both films have the induction electrodes that form two parallel plate capacitors. Driving voltages for the slider are fed through these capacitors that are dedicated for voltage induction.

2.2. *Thrust force in a simple pulse sequence*

This section analyzes thrust force characteristics under a pulse voltage sequence based on the analysis provided in the previous study [19]. The analysis in this paper utilizes an eight-terminal capacitance network model shown in Fig. 2, as well as the previous study. The previous study provided theoretical thrust force for sinusoidal voltages. Since sets of DC voltages that comprises pulse sequences are equivalent to certain phases of sinusoidal wave, thrust force under pulse sequences can be derived by substituting appropriate phases and amplitudes to the thrust force equation for sinusoidal voltages.

With amplitudes and representative phases for stator and slider being V_{st}, V_{sl}, ϕ_{st}, and ϕ_{sl}, the thrust force of the two-to-four phase VITEM is represented as following:

$$f_{\mathrm{ind}} = \frac{\pi C_i C_{m1} V_{sl} V_{st} \{\sin(\theta_x + \phi_{sl} - \phi_{st}) - \sin(\theta_x - \phi_{sl} - \phi_{st})\} - 2\pi C_{m1}^2 V_{st}^2 \sin(2\theta_x - 2\phi_{st})}{(C_i + C_l + C_{sl})p} \quad (1)$$

where θ_x is the slider position in electric angle representation and C_i, C_l, C_{sl}, and C_{m1} are capacitances among electrodes as defined in [19] (see Fig. 2): C_i is the capacitance of induction electrodes; C_l is the capacitance between two phases of the slider electrodes; C_{sl} is the self capacitance of one phase of slider electrodes; C_{m1} is the amplitude of capacitance variation between any phase of slider and any phase of stator electrodes. The induced voltage on the slider was also provided in the previous study and is:

$$V_5 = \frac{C_i V_{sl} \sin\phi_{sl} - 2C_{m1} V_{st} \sin(\theta_x - \phi_{st})}{C_i + C_l + C_{sl}}, \quad V_6 = -V_5. \quad (2)$$

For the first analysis, we assume the voltage sequence depicted in Fig. 3, which is referred to as "sequence A" in the following. In sequence A, the stator's driving electrodes have a set of DC voltages $[+V, 0, -V, 0]$, which is shifted at a certain interval to realize stepwise driving operation. Conversely, a

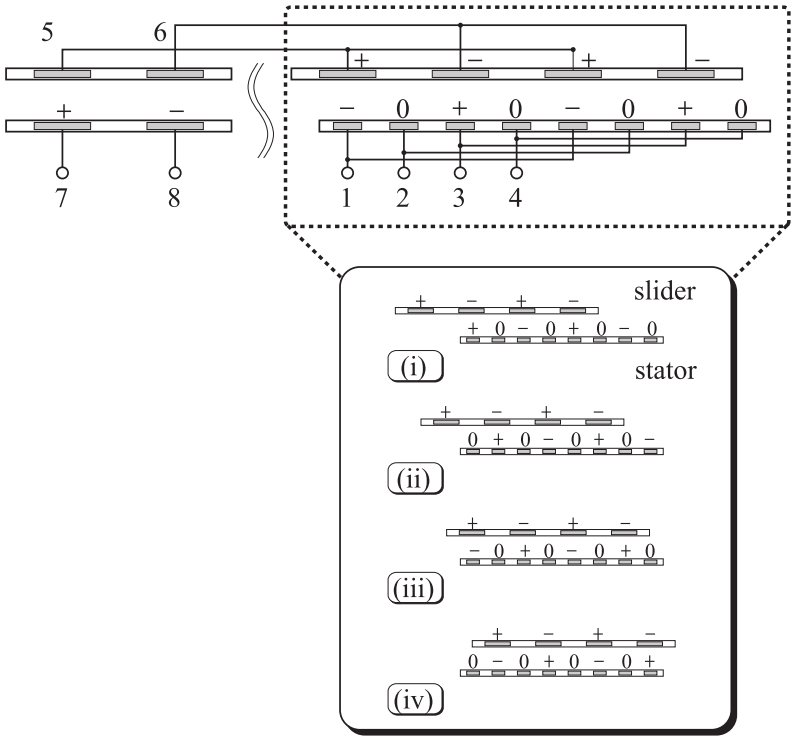


Fig. 3. Voltage sequence A: A basic voltage sequence for driving a two-to-four phase electrostatic motor with voltage induction. A four-phase voltage set of $[+V, 0, -V, 0]$ and a fixed DC voltage set of $[+V, -V]$ are applied to the terminals. Shifting of the four-phase voltage generates four sequential states that are referred to as (i) through (iv).

fixed DC voltage set, $[+V, -V]$ is given for the slider. The resultant voltage sequence has four states that are referred to as (i) through (iv). Since the voltage set $[+V, -V]$ is equivalent to $V[\sin(\pi/2), \sin(3\pi/2)]$ and the set $[+V, 0, -V, 0]$ is to $V[\sin(\pi/2), \sin(0), \sin(-\pi/2), \sin(\pi)]$, substituting $V_{st} = V_{sl} = V$ and $\phi_{st} = \phi_{sl} = \pi/2$ into Eqs (1) and (2) yields slider voltage and thrust force. For example, for the state (i), the induced voltages V_5, V_6 can be derived as

$$V_5 = \frac{V}{C_{\rm i} + C_{\rm l} + C_{\rm sl}}(C_{\rm i} + 2C_{\rm m1}\cos\theta_x), \quad V_6 = -V_5. \tag{3}$$

Thrust force for all the four states are obtained as

$$f_k = -\kappa V^2 \left[C_{\rm m1}\sin(2\theta_x - k\pi) - C_{\rm i}\sin\left(\theta_x - \frac{k-1}{2}\pi\right)\right] \quad (k = 1, 2, 3, 4) \tag{4}$$

where k denotes the state numbers and κ summarizes the coefficients as $\kappa = 2\pi C_{m1}/\{(C_i + C_l + C_{sl})p\}$.

Figure 4 plots normalized thrust forces against slider positions. This particular plot assumes $C_{\rm m1}$: $C_{\rm i}$ = 1:16, which is reported in [19]. The plot shows that, for any fixed voltages, the force characteristics repeat with the cycle of $4p$, which is same as the structural period of the electrodes. For the motor operation, the voltages shift among four states. The resultant slider motion is estimated from the thrust force characteristics. Each thrust force curve has a stable equilibrium point at which the thrust force balances with external force with a negative slope. In case the external force is zero, stable equilibrium

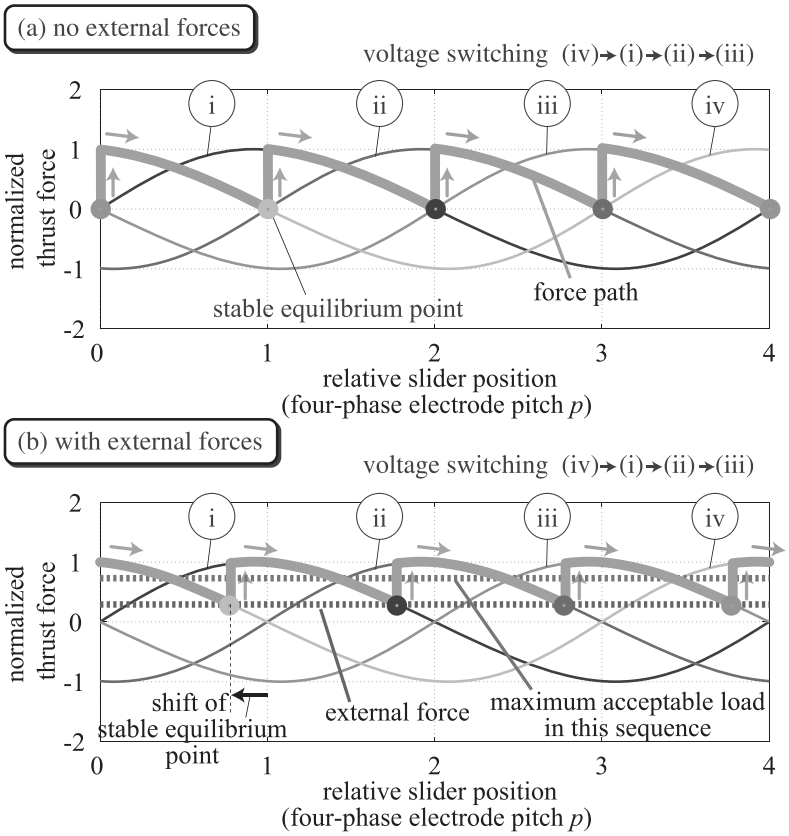


Fig. 4. Normalized thrust force characteristics of voltage sequence A. The numbers from (i) to (iv) correspond to the voltage states in sequence A. The periodic force curves are normalized with the maximum force. (a) Transient thrust force at a driving by sequence A is also depicted in the figure. The force decreases as the slider moves to a stable equilibrium point, then finally halted at the stable equilibrium point. (b) Existence of external force varies the positions of stable equilibrium points.

points can be found on the horizontal axis as shown in Fig. 4(a). By switching the voltages, the slider should jump from one stable equilibrium point to the next one, which results in stepwise motion with the step distance of p. The stable equilibrium points will be shifted if external force, such as friction or load, exists, as shown in Fig. 4(b). Maximum acceptable load, which is the maximum load the motor can continuously drive, can be found at the intersection of the force curves.

Note that these analyses are conducted in static conditions because of the two reasons: most of the important characteristics of the motor such as the maximum force and theoretical stable equilibrium points can be obtained from the static characteristics, and moving time period of each stepping are assumed to be much smaller than pulse period.

2.3. Preliminary experiment

The theoretical discussion about the motion was confirmed by a preliminary experiment. Figure 5 shows dimensions and photos of the motor films used in the experiments throughout this paper. These films were fabricated using flexible printed circuit (FPC) technology. The size of the slider is 95 mm in width and 120 mm in length. The slider's driving electrode measures 86.4 mm by 25 mm, and each

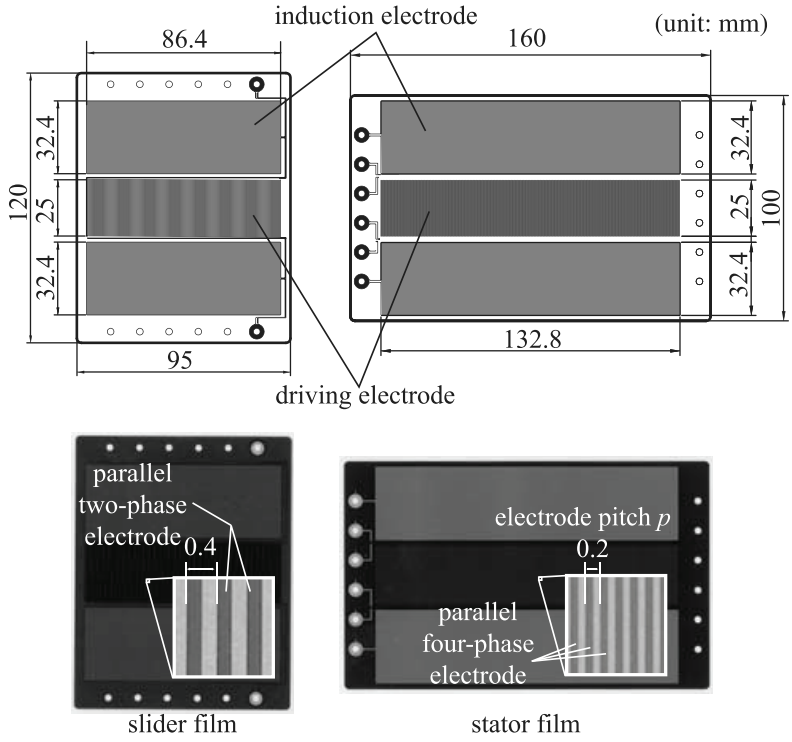


Fig. 5. Schematic diagrams and photos of two-to-four phase voltage-induction motor films. Both films have parallel driving electrode in the middle part of the films and two induction electrodes beside them. The electrode pitches of the driving electrodes were 200 μm in a stator and 400 μm in a slider.

induction electrode measures 86.4 mm long by 32.4 mm wide. The slider is equipped with two terminals to facilitate measurement of the induced voltages on the slider. The size of the stator is 160 mm in width and 100 mm in length. The driving electrode is 132.8 mm by 25 mm and each induction electrode is 132.8 mm by 32.4 mm. The pitches of driving electrodes are 0.4 mm for the slider and 0.2 mm for the stator. The capacitance coefficients for this motor were measured following the method described in [23] and were found as

$$C_{\mathrm{m1}} = 27.5 \text{ pF}$$
$$C_{\mathrm{sl}} = 200 \text{ pF}$$
$$C_{\mathrm{l}} = 85.6 \text{ pF}$$
$$C_{\mathrm{i}} = 464 \text{ pF}.$$

During experiments, glass beads with diameters of 20 μm were scattered between a slider and a stator to reduce frictions. In addition, insulating liquid (FC-77, 3M) was injected between the films to prevent corona discharge between films.

Displacement of the slider was measured by a laser displacement sensor (LC-2400/LC-2440, Keyence). A digital signal processor (ds1104, dSPACE) generated the voltage sequence, which was thousand-fold amplified by six high voltage amplifiers (HVA4321, NF Corporation) before being applied to the motor. The resultant applied voltages were $\pm$ 1 kV. Switching rate of the voltages was either 4, 8, 16, 24, 32, or 40 times/s.

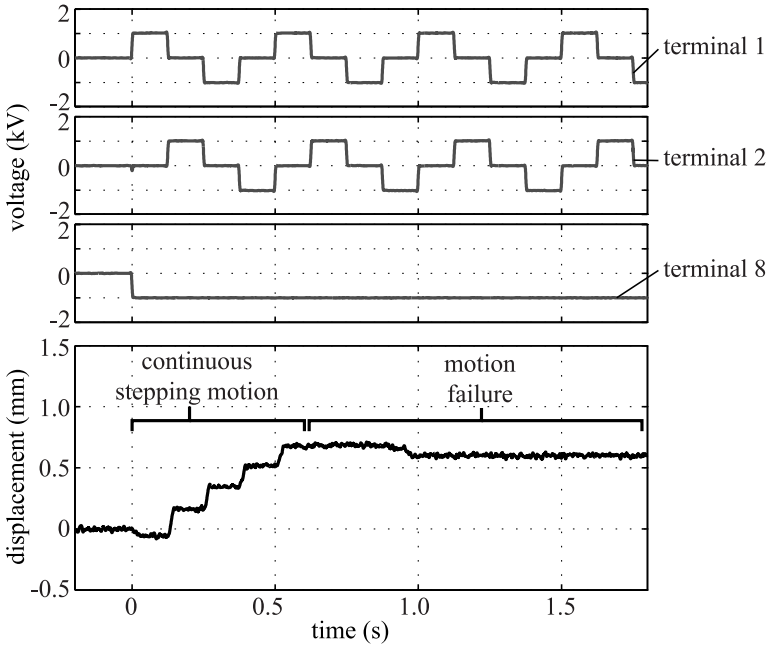


Fig. 6. A motion characteristic by the voltage sequence A. Upper three figures show applied voltages to three terminals. The lower figure shows a motion of the slider. Although the voltages were continuously switched, the motor halted in a few seconds.

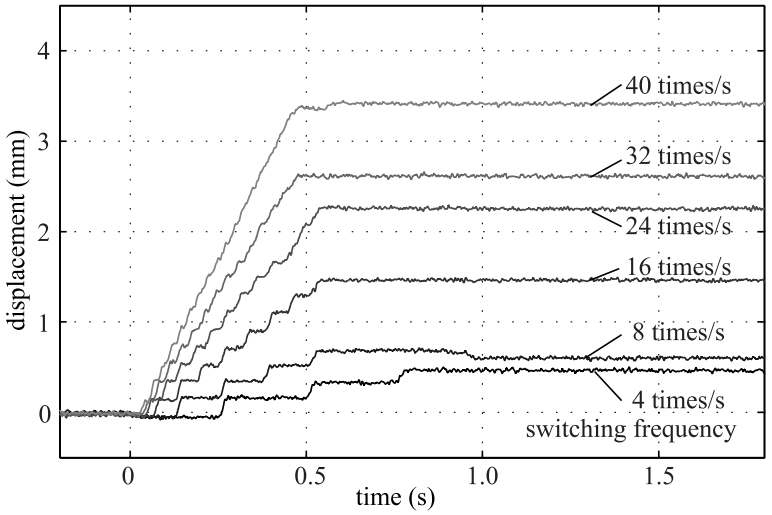


Fig. 7. Motion characteristics by the voltage sequence A with various voltage-switching rates. The motor halted its motions in a short time regardless of switching rates.

Figure 6 plots a measured motion. In the beginning, the motor exhibited stepwise motion with a step size of about 200 μm, which is consistent with the theoretical estimation. After several steps, however, the motor showed unexpected motion and finally halted. Motions at different switching rates are shown in Fig. 7. Regardless of the switching rates, the motor eventually halted after several steps.

This halting behavior is attributed to charge-ups of the insulating layer of the films. It has been reported

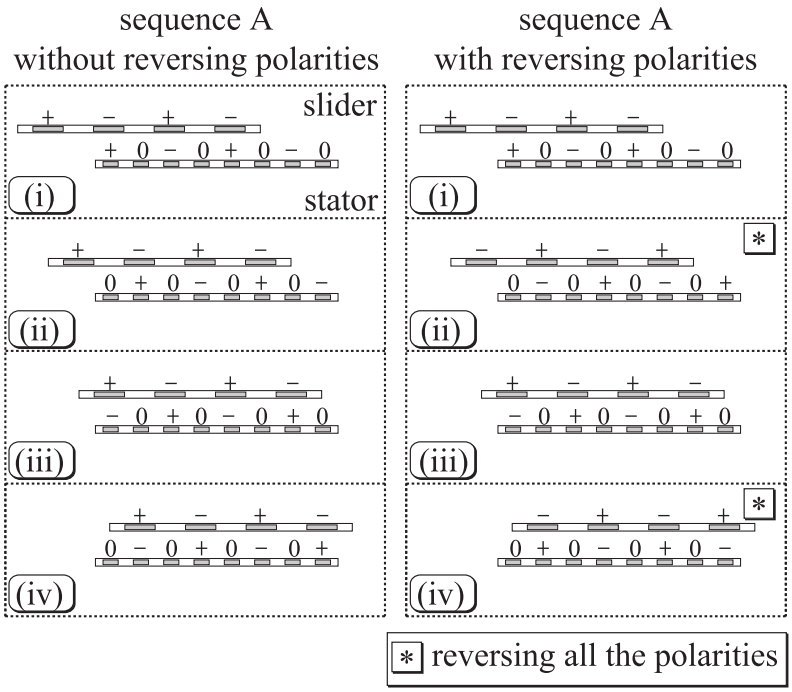


Fig. 8. Modification of voltage sequence A to improve the motion stability. All the voltages are reversed in every two steps.

that applying DC voltages induces surface current on the film surface that accumulates charges on the films [24]. Such charges disturb the electrostatic field within the motor and can lead to the halting behavior. In the tested sequence, the slider voltages were constant which probably caused the motion failure.

3. Proposal of voltage sequence enabling continuous stepping motion

3.1. Modification of sequence A

The halting behaviors would be resolved if the voltage polarities change during operation. This section discusses a modified voltage sequence to improve the stability, which switches two-phase voltages for the slider. The right figure in Fig. 8 is the modified one compared with the last one depicted in the left. In every two steps of the modified sequence, all of the voltage polarities, for both the stator and the slider, are reversed. Since thrust force analyzed in Eq. (4) is proportional to V^2, reversing all the voltages should not change the force characteristics. Therefore, the same motion characteristics can be expected for the modified sequence.

The modified voltage sequence A was evaluated experimentally. Both two-phase and four-phase voltages had amplitude of ± 1 kV. Switching rate of the voltages was 4 times/s. Figure 9 shows the measured motion for the modified sequence. A continuous stepwise motion with a step distance of about 200 μm was observed; no motion failure was found. The result confirmed the improvement of stability by reversing polarities of voltages.

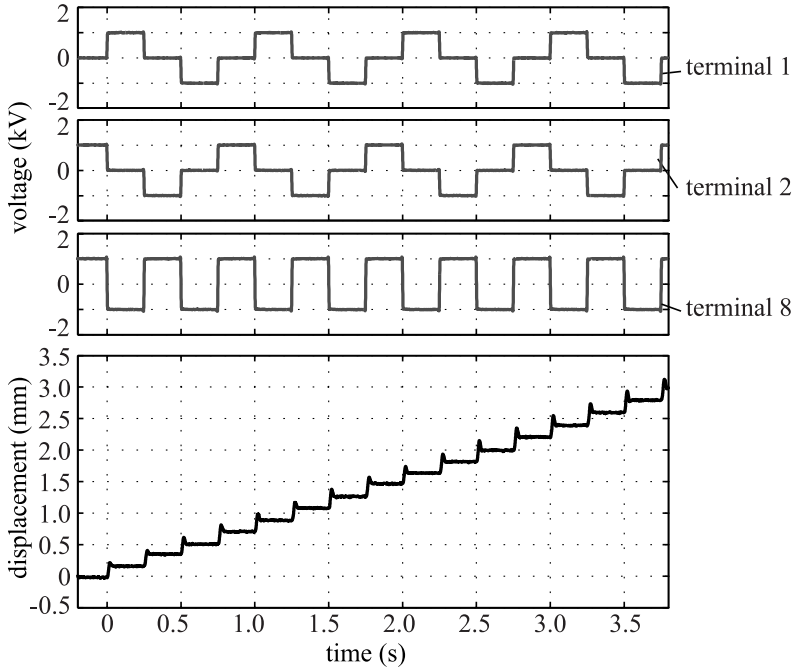


Fig. 9. Motor motion with the modified voltage sequence A. Two-phase voltage switched alternately at the same timing of four-phase voltage switching. A continuous stepping motion is observed.

3.2. Additional voltage sequences

In addition to the sequence A, there are two more possible DC pulse sequences for motor operation. The following discusses the other voltage sequences.

3.2.1. Sequence B

First, we considered a voltage sequence shown in Fig. 10 whose four-phase voltage has a voltage set of $[+V, +V, -V, -V]$. The left figure is the base sequence without reversing polarities, and right one is with the polarity reversing, which can realize continuous stepping motions. Sequence B also creates four states that are different from sequence A and are denoted as states (v), (vi), (vii), and (viii).

Since $[+V, +V, -V, -V]$ is equivalent to $\sqrt{2}V[\sin(3\pi/4), \sin(\pi/4), \sin(-\pi/4), \sin(-3\pi/4)]$, induced voltage, e.g. for state (v), is obtained from Eq. (2) as

$$V_5 = \frac{V}{C_\mathrm{i} + C_\mathrm{l} + C_\mathrm{sl}} \left\{ C_\mathrm{i} + 2\sqrt{2} C_\mathrm{m1} \cos\left(\theta_x + \frac{\pi}{4}\right) \right\}, \quad V_6 = -V_5, \tag{5}$$

and thrust force is calculated as

$$f_k = \kappa V^2 \left[2C_\mathrm{m1} \cos(2\theta - k\pi) + \sqrt{2} C_\mathrm{i} \sin\left(\theta - \frac{k}{2}\pi - \frac{3\pi}{4}\right) \right] \quad (k = 5, 6, 7, 8) \tag{6}$$

where subscript $k(= 5, 6, 7, 8)$ corresponds to the states (v), (vi), (vii), and (viii). Since C_i is much larger than C_{m1}, the second term is dominant in thrust force. Comparison with Eq. (4) suggests that sequence B generates almost $\sqrt{2}$ times larger thrust force than sequence A. This is because sequence B has higher

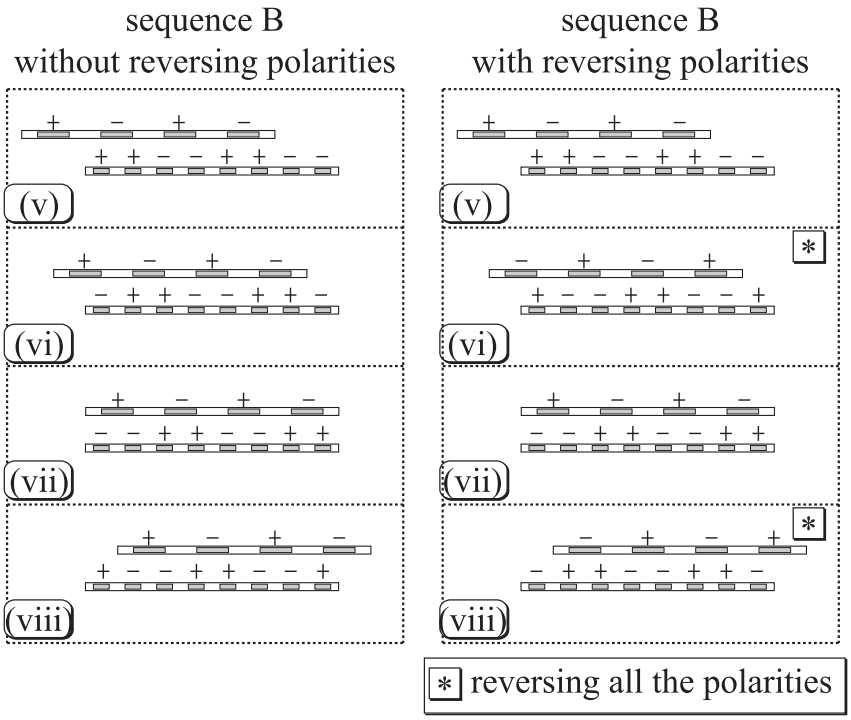


Fig. 10. Voltage sequence B: The voltage sequence with four-phase voltage set of $[+, +, -, -]$. The left figure shows the base sequence without reversing polarities and the right shows the improved sequence that reverses all the voltages in every two steps.

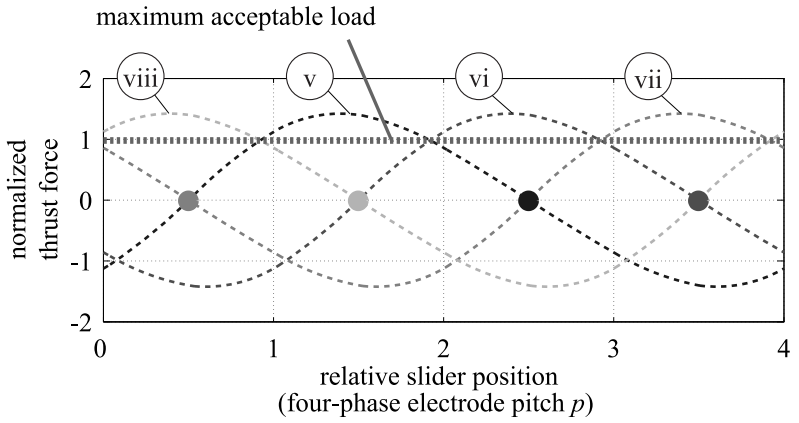


Fig. 11. Normalized thrust force characteristics of voltage sequence B. The numbers from (v) to (viii) correspond to the voltage states in sequence B. The periodic force curves are normalized with the maximum force of the sequence A. Compared with the curves of the sequence A, the maximum values are larger and positions of the stable equilibrium points shift by a half pitch.

voltage in average since the sequence always applies non-zero voltages to all the terminals. Figure 11 plots the force, normalized by the amplitude of force by sequence A. They also form periodic curves, and the four states also generate four stable equilibrium points, which position at the middle of those for sequence A (Fig. 4(a)).

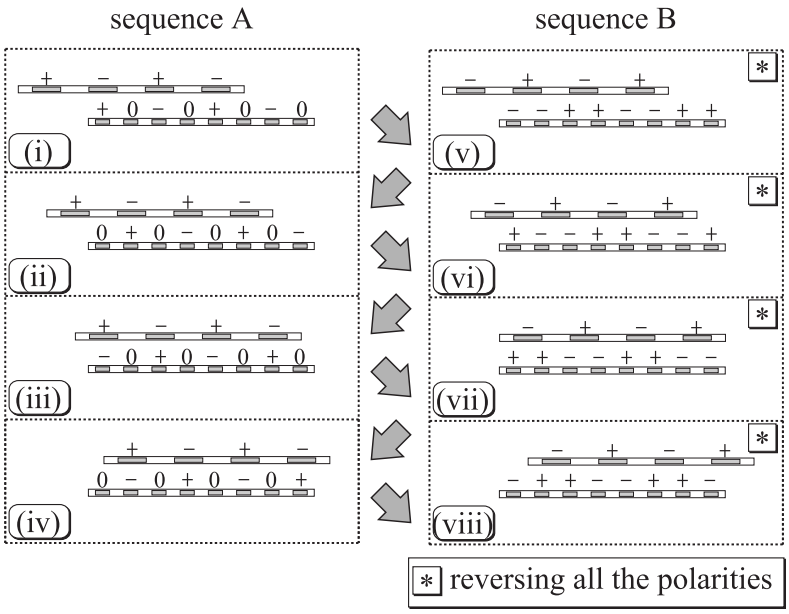


Fig. 12. Voltage sequence C: a sequence that combines sequences A and B.

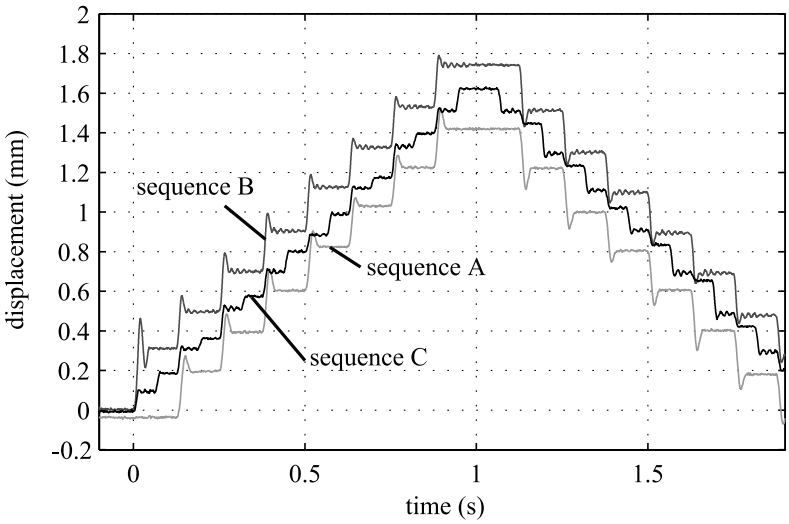


Fig. 13. Motion characteristics under three voltage sequences. Average step widths generated by voltage sequence A and B were 200 μm, and those by voltage sequence C was 70/130 μm depending on voltage states.

3.2.2. Sequence C

Since the positions of stable equilibrium points by sequence A and those by sequence B have a half pitch ($p/2$) difference, their combination can generate eight stable equilibrium points per a structural period and thus can realize finer step distances. Such a combined sequence, referred to as 'sequence C' is shown in Fig. 12. This sequence combines sequence A without polarity reversing and sequence B with reversing for all four states.

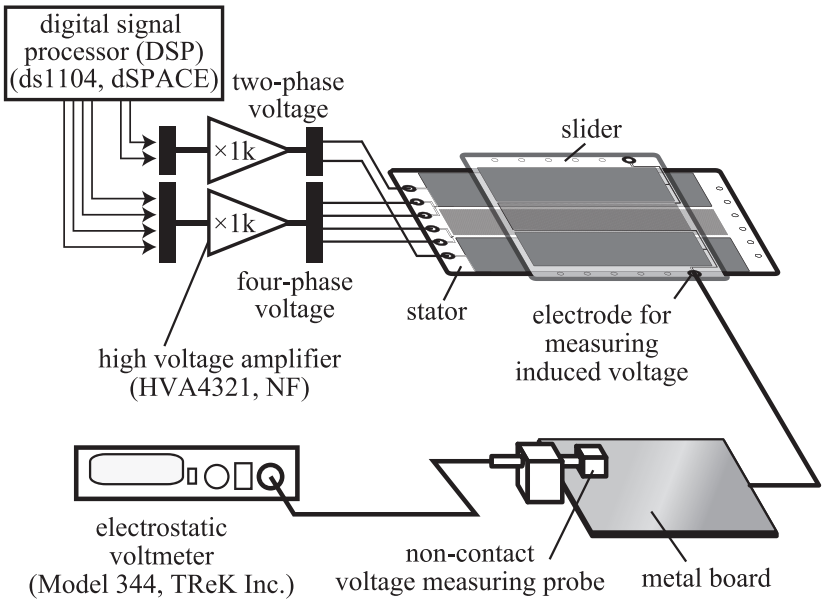


Fig. 14. A setup for measuring an induced voltage on a slider electrode.

4. Experimental evaluation

This section reports the results of our experimental evaluations using the prototype motor that was already described in Section 2.3.

4.1. Motion of the motor

The first set of experiments compared motions of the motor under different voltage sequences A, B, and C, with reversing polarities, were used. Figure 13 shows results of motor displacement measurements. As stated previously, slider showed stepwise motions with step distances of 200 μm for sequences A and B. The step positions in sequence B are located almost at the middle of those in sequence A, as theoretically predicted in Section 2.2. However, a close observation reveals that the step positions of B are not perfectly at the middle of A, but are slightly ahead of the middle points. This was probably be caused by friction between both films. The stable equilibrium points shown in Fig. 4(a) and Fig. 11 are those balancing with zero external loads. When operated under friction force, the stable equilibrium points will shift from the original points as shown in Fig. 4(b), but the shift amounts are different among voltage patterns due to the different gradients of thrust force curves near the stable equilibrium points. Since amplitude of the force curve is larger in sequence B than in sequence A of the same voltage, the equilibrium is reached with a smaller shift. The difference of stepping positions in forward and backward motions can also be attributed to the effect of friction force.

The sequence C, which is a combination of sequences A and B, show almost a half step distances; a slider steps at stepping positions of sequences A and B alternately. The step widths, however, were not constant and were found about 70 μm for voltage shifting from B to A, and about 130 μm for that from A to B, due to the existence of friction. Although step width was not uniform, the results confirmed that voltage sequence C was able to improve positioning resolution.

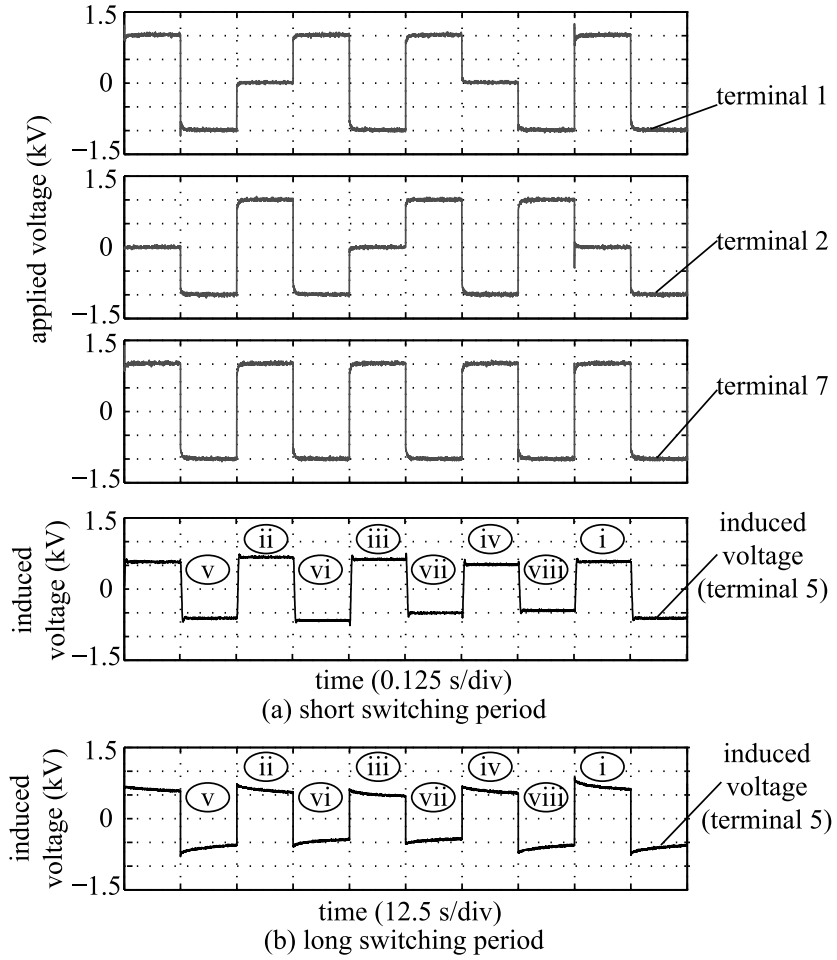


Fig. 15. Induced voltages on an electrode of a slider. Upper three plots in the figure (a) show applied voltages to terminals. The lower plot in the figure (a) and the plot in the figure (b) show induced voltages for a short-period switching and a long-period switching, respectively. Voltage decay is observed in the long-period switching.

4.2. Induced voltage

The second set of experiments measured the induced voltages on the slider. Figure 14 shows the measurement setup. A metal board was connected to one of the slider's electrode to facilitate non-contact measurement using an electrostatic surface voltmeter (Model 344, TReK Inc.). Since the slider's electrodes are electrically floating, non-contact measurement is imperative for this motor. If a contact probe is used, the finite impedance of the probe can cause voltage decay in the low-frequency range, where the motor capacitance has extremely high impedance compared with that of the probe.

Figure 15 shows the measured induced voltage on the slider. Voltage sequence C was used for this experiment since it contains both sequences A and B. The applied voltages were $\pm$ 1 kV, and switching rate was 8 times/s for plot (a) and 0.08 times/s for plot (b). The plot (a) shows that the induced voltages instantly change after shifts of the applied voltages and keep the constant voltages until the next voltage change. The slower switching shown in plot (b), however, shows that the induced voltages gradually decay in a long period. This would be due to small surface currents that run over the surface of the film.

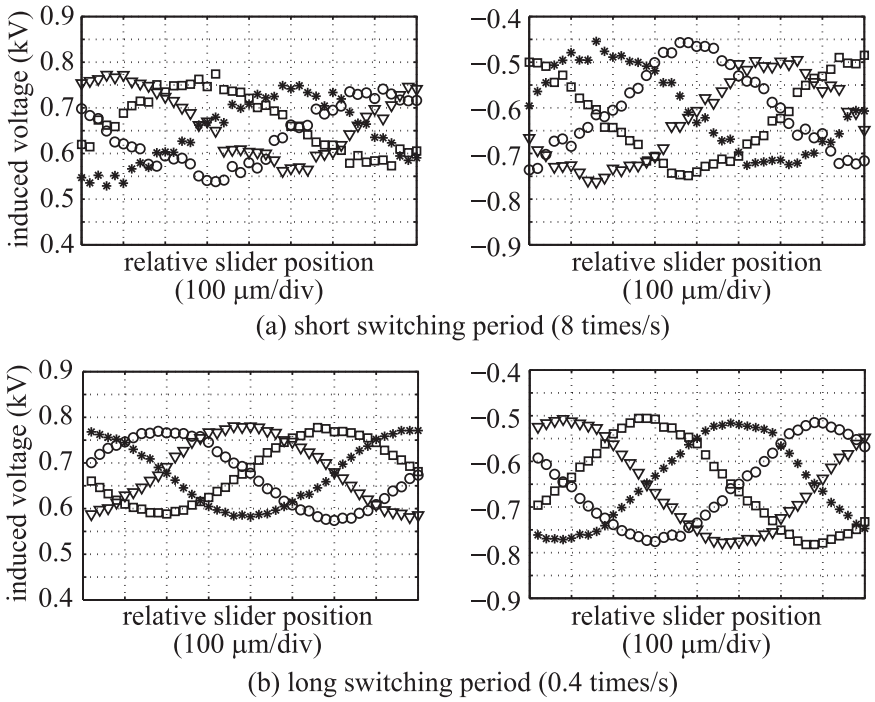


Fig. 16. Relationship between induced voltages and slider position measured for sequence C. Upper figures correspond to a short-period switching and lower figures to a long-period switching. In each of them, the left figure shows the voltages corresponding to sequence A part and the right figure to B part.

As Eqs (3) and (5) suggest, the induced voltages depend on the slider position. Figures 16(a) and (b) plots the peak induced voltages (which are the voltages just after voltage switching and may contain voltage spikes) at each position in different switching periods. The plots on the left column show the voltages that correspond to the sequence A part, and the plots on the right to the B part. As derived in Eqs (3) and (5), the induced voltages are dependent on the slider position and show sinusoidal variations. The average voltages are around $\pm$ 650 V and amplitudes are about 100 V for sequence A part and about 150 V for B part. The corresponding theoretical values are $\pm$ 619 V for average voltages and 73 V and 104 V for amplitudes of A part and B part respectively. The experimental values are slightly larger than the theoretical estimation, which is probably due to the voltage spikes and the charge accumulations that are discussed later.

The dependence of the induced voltages on the switching rate, which was found in Fig. 15, was investigated more in detail as in Fig. 17. Induced voltage of one slider phase was measured at a slider position that maximizes the induced voltage. The figure plots the peak voltages and the minimum voltages after the voltage decays. The decay was probably be caused by the surface current which results in charge accumulations on the surface of the film. The accumulated charges would weaken the electric field and thus decrease the induced voltages. However, once the polarities of the electrode voltages have been reversed, the accumulated charges strengthen the electric field and increase the voltage. Resultantly, in a long switching period in which decay is more significant, the peak voltages became higher, while the minimum voltages became lower. These increases of voltages after switching can strengthen the thrust force.

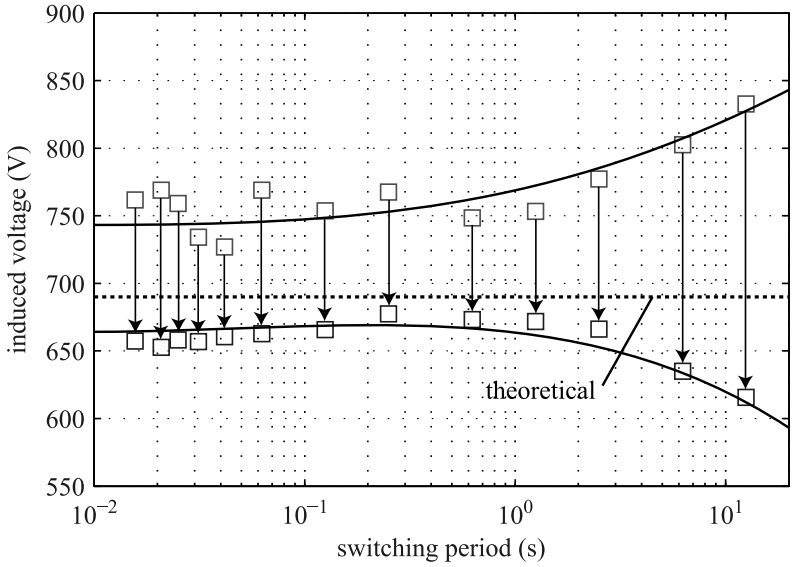


Fig. 17. Decay of induced voltages. The plot shows the peak and the minimum voltages during one switching period. The dashed horizontal line represents the theoretical value.

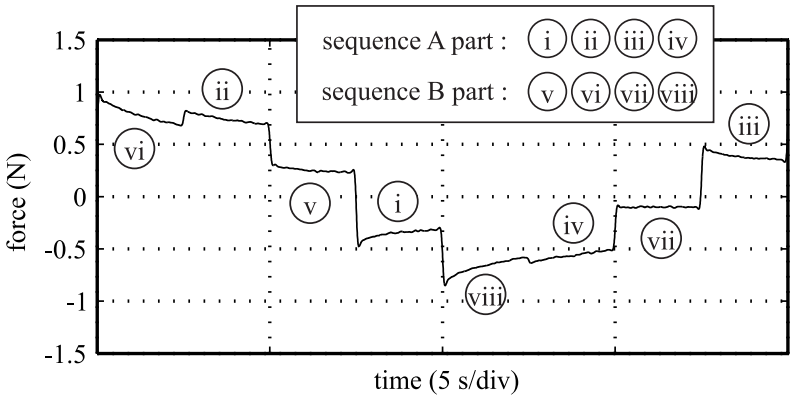


Fig. 18. A thrust force variation over time in sequence C. The slider position was fixed during the measurement. It can be seen that the thrust force instantaneously changed after the switching of pulse voltages and then decayed.

4.3. Force measurement

The final set of the experiments measured the thrust force to reveal the motor output performance. For force measurement, the slider was connected to a load cell (LTS-1KA, Kyowa Electronic Instruments). Dynamic strain amplifiers (DPM-711B, Kyowa Electronic Instruments) amplified signals from the load cell to obtain thrust force. The position of the slider was adjusted by a linear stage and was fixed during each measurement. The force measurement utilized voltage sequence C, which gives force characteristics of sequences A and B in one measurement.

Figure 18 shows a variation of thrust force over time, at a fixed slider position. The measurement observed decay of thrust force in each state, which was probably be caused by the decay of the induced voltages.

Table 1
Comparison of force amplitudes

	Seq. A	Seq. B
Theoretical	0.54 N	0.82 N
Measured (long switching period)	0.89 N	0.98 N
Measured (short switching period)	0.43 N	0.45 N

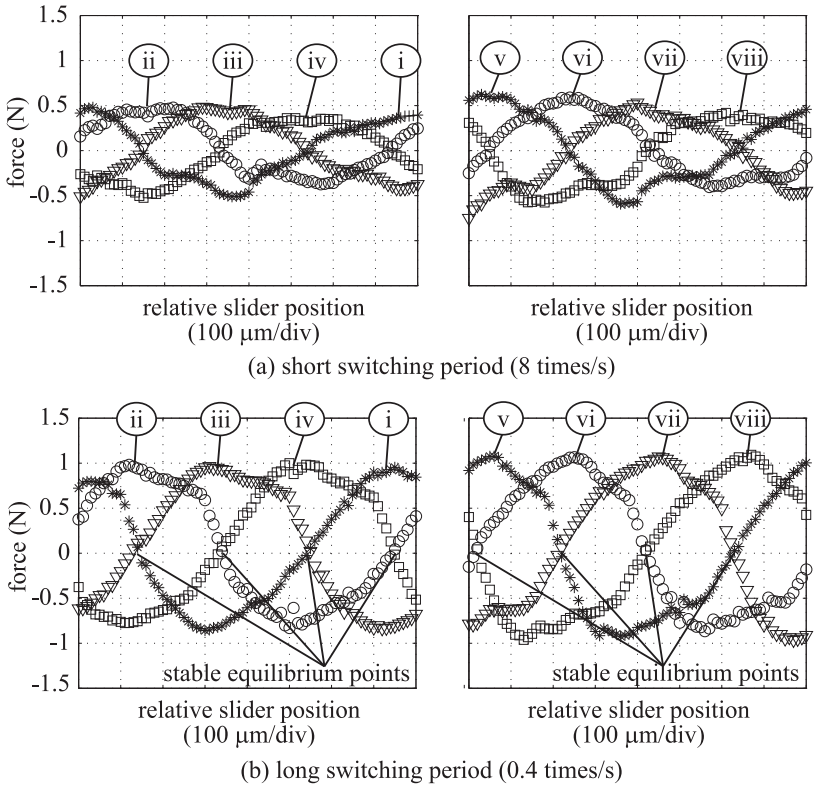


Fig. 19. Relationships of thrust forces and slider position in sequence C. The plots show peak thrust forces before force decay. The upper plots are for a short switching period and the lowers are for a long period. Left plots correspond to sequence A part and right plots to B part.

The same measurements were carried out at different slider positions and with different switching rates. The measured results are summarized in Fig. 19. The plots show the peak thrust force of each state at different slider positions. Switching rates were 8 times/s and 0.4 times/s in plots (a) and (b), respectively, and plots on the left show forces corresponding to sequence A, and plots on the right correspond to B. Each state forms a quasi-sinusoidal curve that corresponds to the theoretical estimations plotted in Figs 4 and 11. The results also indicate that the stable equilibrium points for sequence B shift about a half electrode pitch from those for sequence A.

Table 1 compares the measured thrust forces with theoretical estimations in terms of force amplitude. The theoretical estimations are based on Eqs (4) and (6) that do not consider force decay. Both in theory and measurements, forces in sequence B are larger than those in sequence A. Their ratios, however, are smaller in measurements. The reason for this is not clear and will be investigated in future. Due to the decay of the thrust force and resultant rise of the thrust force shortly after voltage switching, the thrust

forces for longer switching periods show much larger values than those for shorter periods. Still, the measured values and theoretical one can be found within a similar range, which validates the theoretical calculations.

5. Conclusions

This paper revealed appropriateness of a pulse voltage operation for a two-to-four phase VITEM by both analyses and experiments. Appropriate voltage sequences are proposed and the characteristics and transient charge effects for force enhancement are demonstrated.

First, the force analysis and a preliminary experiment using a simple pulse voltage sequence revealed the possibility of stepping actuations, but it also revealed halting motion behavior probably due to charge accumulations. To realize continuous stepping actuations, the paper proposed an improved voltage sequence, in which voltage polarities reverse every time to suppress the charge accumulations. The paper also proposes two more variants of the voltage sequences, all of which exhibited continuous stepping actuations.

The detailed experimental investigations validated theoretical estimations about induced voltages and thrust forces. The experimental results also observed instant enhancement and following decay of induced voltages and thrust forces. Those effects would be caused by accumulation of the charges on motor films. Although the charges prevented the motion in preliminary experiment, we revealed that voltage switching can use the charges for instant enhancement of forces. The enhancement can be effective for a driving with high load and static friction.

The further experiments about the transient characteristics revealed the relations between the motor performance and switching rate of pulse voltages. In addition, the motor successfully demonstrated continuous stepping motions at any tested switching rates.

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